

UrjaMitra AI System Architecture

END-TO-END DIGITAL TWIN PIPELINE



PROCESSING
Zero Cloud Latency



INSTALLATION
100% Non-Invasive



Clip-on Current Sensors

Snap-on split-core sensors capture exact machine power draw in real-time without cutting any factory wires.

SCT-013 NON-INVASIVE



Machine Vitals Sensors

Magnetic thermal and vibration nodes attach directly to motor chassis to detect mechanical stress.

MPU6050 VIB DS18B20 TEMP

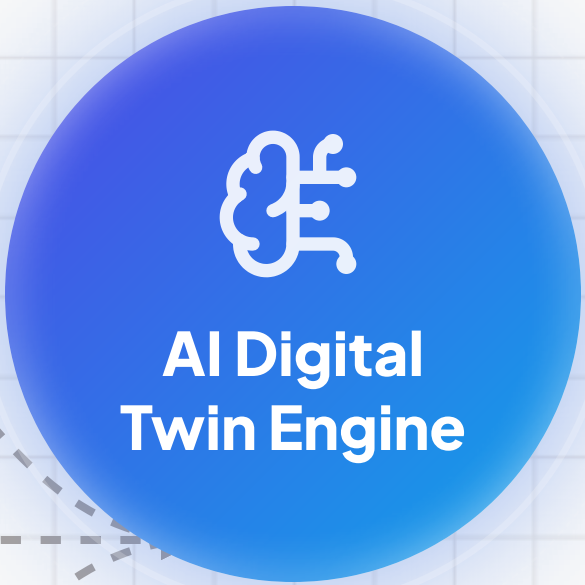


IoT Edge Hub

Cleans electrical noise locally and packages telemetry data for ultra-secure transmission via factory Wi-Fi.

ESP32 MQTT SECURE

THE BRAIN



AI Digital
Twin Engine

PROCESSING STACK

- ✓ Timeseries Database
- ✓ ML Power-Waste Rules
- ✓ Predictive Maintenance



Financial Dashboard

Translates raw wattage into exact Rupee (₹) savings. Owners can visually see idle machine power bleeding in real-time.

NEXT.JS WEB APP LIVE CHARTS



Proactive Alerts

Instantly triggers Push Notifications or SMS when a motor's heat/vibration predicts a catastrophic failure next week.

PUSH API SMS GATEWAY



ZED Green Audit

Automatically tallies Carbon (CO2) reduction metrics, generating compliant reports instantly for MSME ZED Certification.

PDF GENERATOR ZERO DEFECT